

Christina Wang

Lederman Fellow at Fermilab

617-997-1187 ·
christiw@fnal.gov

PROFESSIONAL EXPERIENCE

Lederman Fellow at Fermilab
Postdoctoral Scholar Research Associate at Caltech

September 2024 – present
April 2024 – September 2024

EDUCATION

California Institute of Technology
Ph.D. in Physics. Advisor: Maria Spiropulu
Thesis: [Search for New Physics with the CMS Experiment and QIS-enabled Technology](#)

October 2018 – March 2024

Massachusetts Institute of Technology
Bachelor of Science in Physics

September 2015 – June 2018

FELLOWSHIPS AND AWARDS

APS Mitsuyoshi Tanaka Dissertation Award in Experimental Particle Physics	2026
CMS Thesis Award	2024
DOE Graduate Student Research Fellowship (SCGSR)	2022
2022 Snowmass Poster Award	2022
Caltech Rosenbaum-Faber Family Fellowship	2018

APPOINTED POSITIONS

CMS Long-lived Particle Subgroup Convener 2024 – 2026
Reviewing and managing 20+ active long-lived particle analyses. 17 public results during tenure.

CMS Exotica Methods Forum Contact 2023 – 2024
Started monthly meeting, documentation website, and common EXONanoAOD data format to foster exchange and preservation of unconventional techniques unique to exotica searches

GRANTS AND FUNDING

Fermilab LDRD Superconducting Nanowire Detectors for High-Energy Particle Detection	2025
Fermilab Detector R&D New Initiatives Superconducting Nanowire Detectors for High-Energy Particle Detection	2024

RESEARCH EXPERIENCE

CMS Experiment 2017 – present
Searches for Long-lived Particles (LLPs) Decaying in the CMS Muon Detectors

- Discovered muon detector shower (MDS) signature and developed MDS reconstruction and identification techniques as a powerful tool for LLP searches
- Led the first LLP searches using MDS, achieving the best sensitivity for LLPs with proper decay lengths from 5–10000 m [7], [9]
- Contributed to the search for heavy neutral leptons using the MDS signature [8]
- Designed and led the analysis team on the search for LLPs produced from B meson decays using the MDS signature [6]

- Designed and developed the MDS trigger for Run 3, improving trigger efficiency by an order of magnitude [4], [19], [20]
- Led an analysis team to analyze 2022–2024 data with the new MDS trigger [18]

Reinterpretation of CMS MDS Search to New Physics Models

- Developed the [new Delphes module for MDS](#) to enable future reinterpretations based on the parameterized detector response function measured for the first CMS endcap MDS search
- Recast the first CMS endcap MDS search, showing its complementarity to other LLP experiments for many LLP models [10], [13]

Search for Heavy Stable Charged Particles (HSCP)

- Search for HSCP with highly ionizing signature, providing stringent limits on the masses of gluinos up to 2 TeV and tau sleptons up to 0.69 TeV [14]
- Studied sensitivity of an additional double-HSCP analysis category and calculated systematic uncertainties of key variables and trigger efficiency scale factor

Search for LLPs Using Out-of-time Trackless Jets

- Provides the best sensitivity for long-lived neutralinos with proper decay lengths of 0.3–1.0 m, excluding masses up to 1.18 TeV [12]
- Developed the key discriminating variables used as inputs to the DNN for LLP-jet tagging

Precision Timing: MIP Timing Detector (MTD) for HL-LHC Upgrade

- Measured light yield of LYSO with different LYSO+SiPM geometries, set up the data acquisition system, and prepared hardware for test beams at FNAL [15]
- Performed test beam measurements to test the TOFHIR ASIC readout chip and radiation hardness of LYSO crystals at FNAL and LANL [11]

Measurements of Differential Z Boson Production Cross Section

- Measured Drell-Yan differential cross section with focus on the low- p_T region [16]
- Optimized differential bins, performed statistical analysis and unfolding procedure, and presented findings at the CMS SMP (Standard Model Physics) group

Dark Matter and Quantum Sensing Experiment

2021 – present

R&D for Superconducting Nanowire Single Photon Detectors (SNSPDs)

- Developed and built an integration and characterization system to benchmark the performance of large-area (mm^2) SNSPDs, including full cryogenic and room-temperature infrastructure
- Measured an unprecedentedly low dark count rate of 10^{-3} Hz for mm^2 SNSPDs [1]
- Characterized the SNSPD response to high-energy particles for the first time at the Fermilab Testbeam Facility [3] and followup study at the CERN SPS Testbeam facility [2]

BREAD: Broadband Reflector Experiment for Axion Detection

- Axion search using a parabolic reflector to focus axion-converted photons onto SNSPDs, providing the best sensitivity in the meV–eV mass range

Dark Matter Detection with GaAs + SNSPDs

- Building a prototype to search for sub-GeV dark matter candidates by coupling SNSPDs with gallium arsenide (GaAs), a bright cryogenic scintillator
- Developed an x-ray calibration system for the detector prototype with a Co57 source and high-purity germanium detectors to detect Compton-scattered photons

- Benchmarked performance (time jitter, synchronization, and noise metrics) of various commercial digitizers for the data acquisition system to inform procurement decisions

PUBLICATIONS & PUBLIC RESULTS

Peer reviewed publications:

1. C. Wang et al. "Temperature-Dependent Characterization of Large-Area Superconducting Microwire Array with Single-Photon Sensitivity in the Near-Infrared." [JINST 20 P08005](#)
2. C. Wang et al. "Towards high-efficiency particle detection using superconducting microwire arrays." [JINST 20 P12033](#)
3. C. Wang et al. "High Energy Particle Detection with Large Area Superconducting Microwire Array." [JINST 20 P03001](#)
4. C. Wang et al. "High Multiplicity Trigger for Long-Lived Particles in CMS detector." [Accepted by JINST](#)
5. CMS Collaboration. "Search for vector-like leptons with long-lived particle decays in the CMS muon system in proton-proton collisions at $\sqrt{13}$ TeV." [JHEP 08 \(2025\) 156](#)
6. CMS Collaboration. "Search for b-hadron decays to long-lived particles in the CMS endcap muon detectors." [Phys. Rev. D 113, 012009 \(2026\)](#)
7. CMS Collaboration. "Search for long-lived particles decaying in the CMS muon detectors in proton-proton collisions at $\sqrt{13}$ TeV." [Phys. Rev. D 110, 032007 \(2024\)](#)
8. CMS Collaboration. "Search for long-lived heavy neutral leptons decaying in the CMS muon detectors in proton-proton collisions at $\sqrt{s} = 13$ TeV." [Phys. Rev. D 110, 012004 \(2024\)](#)
9. CMS Collaboration. "Search for long-lived particles decaying in the CMS endcap muon detectors in proton-proton collisions at $\sqrt{13}$ TeV." [Phys. Rev. Lett. 127, 261804 \(2021\)](#)
10. C. Wang et al. "Energetic long-lived particles in the CMS muon chambers." [Phys. Rev. D 108, 055040 \(2023\)](#)
11. C. Wang et al. "Radiation damage in 20-cm long LYSO:Ce and BaF₂:Y crystals." [IEEE Transactions on Nuclear Science, vol. 71, no. 9, 2024](#)
12. CMS Collaboration. "Search for long-lived particles using out-of-time trackless jets in proton-proton collisions at $\sqrt{13}$ TeV." [JHEP 07 \(2023\) 210](#)
13. C. Wang et al. "Long-lived heavy neutral leptons with a displaced shower signature at CMS." [JHEP 02 \(2023\) 011](#)
14. C. Wang et al. "Search for heavy long-lived charged particles with large ionization energy loss in proton-proton collisions at $\sqrt{s} = 13$ TeV." [JHEP 04 \(2025\) 109](#)
15. The CMS MTD Collaboration. "Test beam characterization of sensor prototypes for the CMS Barrel MIP Timing Detector." [JINST 16 \(2021\) P07023](#)
16. CMS Collaboration. "Measurements of differential Z boson production cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV." [JHEP 12 \(2019\) 061](#)
17. S. Xie et al. "Entangled Photon Pair Source Demonstrator using the Quantum Instrumentation Control Kit System." [IEEE Journal of Quantum Electronics, vol. 59, no. 5, 2023](#)

Public results:

18. CMS Collaboration. "Search for long-lived particles with a pair of muon detector showers in proton-proton collisions at $\sqrt{s} = 13.6$ TeV" CMS Physics Analysis Summary [CMS-PAS-EXO-25-005](#), 2025.
19. CMS Collaboration. "Performance of long lived particle triggers in Run 3." CMS Detector Performance Summaries [CMS-DP-2023-043](#), 2023.
20. CMS Collaboration. "CSC High Multiplicity Trigger in Run 3." CMS Detector Performance Summaries [CMS-DP-2022-062](#), 2022.
21. Alimena et al. "Review of opportunities for new long-lived particle triggers in Run 3 of the Large Hadron Collider." [arXiv:2110.14675](#)

SERVICE

Meeting of the Division of Particles and Fields of APS

- Instrumentation track convener and session chair

2026

CMS Exotica Workshop	2024 – 2025
• Long-lived session co-organizer and co-chair	
CMS Data Analysis School	2024 – 2026
• Created and led the first two-day data analysis exercise focusing on long-lived particles	
• Introduce groups of 10 graduate students to data analysis with LHC data	
Co-organizer of Fermilab Research Techniques Seminar	2026 –
Journal referee for papers in Physical Review Letters	2025 –

OUTREACH

Saturday Morning Physics	2025 – present
• Co-organizer for a biannual 11-lecture series aimed at high school students	
• Lecturer for “Particle Physics at the LHC”	
CU*IP at Fermilab	2024 – 2025
• APS Conferences for Undergraduate Women and Gender Minorities in Physics	
• Presenter, session chair, and member of the local organizing committee	
Fermilab YouTube Shorts and Instagram Reels for World Quantum Day	2023
Interview with Symmetry Magazine on LLP Search	2021
Caltech FUTURE of Physics Program	2018 – 2022
• An annual two-day program that brings undergraduate women in physics to Caltech for mini-workshops on navigating graduate applications and graduate student life, particularly as a woman	
• Served as a panelist, gave lab tours and presentations, and starred in a five-minute video in the pandemic edition of the program introducing my career path and student life at Caltech	
MIT Educational Councilor (Greater LA Region)	2018 – 2022
• Conducted in-depth interviews of about 10 first-year undergraduate applicants per year and submitted reports to the MIT Admissions Office	
MIT Code for Good (Co-Director)	2015 – 2018
• Organized an annual one-month January-term class, Code for Good, matching teams of 3–4 students with non-profit organizations for programming projects	
• Organized lectures, office hours, and final project expositions, and secured sponsorships from Facebook, Microsoft, and Yahoo to fund the program	
MIT Physics Tutor	2017
• Tutored two students on Special Relativity for 5 hours per week	
MIT Physpop Program, Co-treasurer	2016
• Organized a week-long Freshman orientation program in Physics for 50 incoming students	
• Coordinated introductory lectures from MIT physics professors and lab tours for incoming freshmen	

SEMINARS

Search for New Physics with CMS and Quantum Sensors LBNL	October 2023
Search for Neutral Long-lived Particle Decaying in the CSCs UNM NUPAC Seminar (Virtual)	October 2021

TALKS & POSTERS

CMS Experiment

Searches for Long-lived Particles and Unconventional Signatures with CMS

International Conference on High Energy Physics (Natal, Brazil) August 2026

Search for Vector-like Leptons and Vector-like Quarks in CMS

Moriond Electroweak Interactions & Unified Theories (La Thuile, Italy) March 2025

Long-lived Particles in the Muon Detectors

14th LHC LLP Workshop (Tokyo, Japan) July 2024

Proton-Induced Radiation Damage in 20 cm Long LYSO:Ce and BaF₂:Y Crystals (Poster)

2023 IEEE NSS MIC RTSD (Vancouver, Canada) November 2023

Search for New Physics with Long-lived and Unconventional Signatures in CMS

SUSY2023 (Southampton, UK) July 2023

Invisible Higgs Decay at CMS

Standard Model at the LHC Workshop (Batavia, IL) July 2023

Status and Prospect of Search for LLP at ATLAS and CMS

Mitchell Conference (College Station, TX) May 2022

Reinterpretation for Neutral Long-lived Particle Decaying in the CSCs

LHC Reinterpretation Forum (CERN) December 2022

12th LHC LLP Workshop (Virtual) November 2022

APS April Meeting (New York, NY) April 2022

10th LHC LLP Workshop (Virtual) November 2021

RAMP Seminar Series (Virtual) September 2021

Search for Neutral Long-lived Particle Decaying in the CSCs

Poster at Snowmass Summer Workshop (Seattle, WA) July 2022

9th LHC LLP Workshop (Virtual) May 2021

APS April Meeting (Virtual) April 2021

Dark Matter and Quantum Sensing Experiment

High Energy Particle Detection with Large Area Superconducting Microwire Array

Applied Superconductivity Conference 2026 (Pittsburgh, PA) September 2026

APS Joint March Meeting and April Meeting (Denver, CO) March 2026

CPAD (Philadelphia, PA) November 2025

Annual Meeting of the International Muon Collider Collaboration (Hamburg, Germany) May 2025

CPAD (Nashville, TN) November 2024

APS Joint March Meeting and April Meeting (Anaheim, CA) March 2025

BREAD: Broadband Reflector Experiment for Axion Detection (Invited)

Applied Superconductivity Conference 2024 (Salt Lake City, UT) September 2024

Searching for Axions and Dark Photons with SNSPDs in the BREAD experiment

CPAD (Nashville, TN) November 2024

CPAD (Menlo Park, CA) November 2023

APS April Meeting (Minneapolis, MN) April 2023

CPAD (Stony Brook, NY) November 2022

FPGA Based Signal Generation and Readout of SNSPD Signals for Quantum Communication

APS March Meeting (Las Vegas, NV) March 2023

SUPERVISION AND MENTORSHIP

Ph.D. Students

- **Matias Barria** (Purdue). Higgs to Di-muon with CMS Run 3 data (2025–present).
- **Elise Sledge** (Caltech). Higgs to Di-muon with CMS Run 3 data (2025–present).
- **Ludovico Mori** (Caltech). High energy particle detection with SNSPD (2024–present).

- **Thomas Sievert** (Caltech). High energy particle detection with SNSPD (2024–present).
- **Alexander Albert** (Caltech). Search for LLPs with MDS using Run 3 data (2024–present).
- **Chang Sun** (Caltech). Search for LLPs with MDS using Run 3 data (2024–present).
- **Paul Simmerling** (Caltech). Search for LLPs with MDS using 2022 data (2022–2023).
- **Anthony Aportela** (UC San Diego). Search for LLPs with MDS using B-Parking dataset (2022–2023).
- **Aram Hayrapetyan** (Yerevan University). Search for LLPs with MDS using B-Parking dataset (2022–2024).

Masters Students

- **Valentina Vega** (Universidad Técnica Federico Santa María). Building cosmic muon veto for SNSPDs (2024–2025).
- **Claudio San Martín** (Universidad Técnica Federico Santa María). High energy particle detection with SNSPD (2024).

Undergraduates

- **Joseph Sorel** (University of Chicago). Optimizing time resolution of large area SNSPDs (2026 Summer).
- **Jiahui Xie** (Caltech). Studied the impact of different muon reconstruction algorithms for Higgs to di-muon. (2025 Summer).
- **Sherry Wang** (Caltech WAVE Fellow). Reconstruction of boosted di-jet from Higgs boson decay into four bottom quarks (2023 Summer).
- **Ariela Yashinsky** (URA-Fermilab: Undergraduate Women in STEM). Reconstruction of boosted di-jet from Higgs boson decay into four bottom quarks (2023 Summer).
- **Samantha Contreras** (Caltech WAVE Fellow, went on to UCLA PhD). Phenomenology study of MDS from LLPs in the dark shower model; Goldwater Scholarship and Caltech Vodopia-Hasson Poster Competition winner (2022 Summer).
- **Gabby Dituri** (Caltech). Reconstruction of low-mass LLPs in the CMS muon system; poster at the National Collegiate Research Conference at Harvard (2021) (2019–2022).
- **Nathan Suri** (Caltech, went on to Yale PhD). Phenomenology study to search for quirks with the CMS tracker and MIP Timing Detector at HL-LHC; talk at the [11th LHC LLP Workshop](#) (2018–2022).
- **Dawn Tang** (Caltech, went on to Berkeley PhD). CMS MTD upgrade: measured and studied the light collection efficiency of scintillation crystals with different geometries and wrapping materials (2018–2019).
- **Linus Kim** (Caltech, went on to Berkeley PhD). CMS MTD upgrade: automated power supply settings and the data acquisition and logging procedure for test beams at FNAL (2018–2019).